

Service Learning Report

Project Haalu - A Study of Milk Types and Their Correlation with
Quality Variables

*Submitted in Partial Fulfillment of the Requirements for the Award of the
degree in Bachelor of Science in Economics, Mathematics and Statistics*

by

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Chapter 1

Introduction

Service learning is an educational approach that combines academic learning with community service, allowing students to apply their knowledge and skills to real-world challenges while fostering civic responsibility. Unlike traditional volunteering, service learning integrates structured reflection and academic objectives, so that students not only contribute to their communities but also gain deeper insights into societal issues, critical thinking skills, and personal growth. This experiential learning method benefits both students and the community, creating a mutually enriching relationship that enhances education through hands-on engagement.

A key component of service learning is reflection, where students analyze their experiences, draw connections between their service and academic learning, and explore the impact of their contributions. This reflective process deepens their understanding of social issues, promotes civic engagement, and encourages a lifelong commitment to community involvement.

Service learning can take many forms, such as tutoring underprivileged students, conducting environmental conservation projects, working with nonprofit organizations, or developing public health initiatives. It benefits the students, who gain hands-on experience, leadership skills, and a deeper appreciation for civic responsibility, and the communities they serve, who receive practical support and locally relevant solutions.

1.1 Aim and Objectives

Milk is a widely consumed nutritional staple, valued for its content of macronutrients, vitamins, and minerals. Its composition and quality vary depending on the source, processing method, and external factors such as feeding practices and storage conditions. This report considers four milk sources, **cow, buffalo, goat, and mixed milk** (a blend of different sources), each of which differs in fat content, protein levels, pH, and other physicochemical properties that influence nutritional value and stability. Buffalo milk in particular is known from the literature to differ substantially from cow milk, most notably in having a markedly higher fat and protein content [12], and its inclusion here is intended to make the source-wise comparison more representative of milk actually sold and consumed in the Indian market.

This report analyzes these milk types and their relationship with key quality variables, including fat percentage, solids-not-fat (SNF), protein content, pH, and the presence of added water, using a dataset constructed for this study (see Chapter 2 for details on its provenance). Examining these relationships helps assess milk purity, nutritional ad-

equacy, and the possibility of adulteration, and provides a starting point for connecting these patterns to what published dairy-science literature reports about spoilage and processing.

Milk quality assessment matters both for consumer health and for regulatory compliance within the dairy industry. This report’s own contribution is a descriptive and correlational analysis of a constructed dataset; questions that would require laboratory measurement over time (for example, microbial growth rates or the direct effect of refrigeration) are addressed here through the literature review rather than through original experimentation, and the objectives below are written to reflect that scope honestly.

The main objectives of this study are:

- To construct a representative, literature-grounded dataset of physicochemical milk-quality variables across cow, buffalo, goat, and mixed milk, and to document its provenance and construction method transparently.
- To analyze the correlation between milk quality indicators, fat, SNF, protein, pH, and added-water percentage, and to examine how they move together within the dataset.
- To compare the nutritional and physicochemical properties of the four milk sources (cow, buffalo, goat, and mixed) to identify relative differences in composition.
- To review, through the literature, how microbial activity, storage temperature, and processing techniques (pasteurization, homogenization, refrigeration) are reported to affect milk spoilage and shelf life, and to relate these findings qualitatively to the patterns observed in the dataset.
- To explore, as a supplementary exercise, whether a simple classification model (Random Forest) can distinguish the dataset’s Standard / Low Quality / Fermented labels from the recorded physicochemical variables.
- To offer data-informed, appropriately hedged suggestions for milk storage, adulteration detection, and processing practice, clearly distinguishing what the dataset shows from what the literature reports.

1.2 Literature Review

Milk composition differs meaningfully across species and production systems, which in turn affects nutritional value and processing behaviour. A comparative study of sheep and goat milk found pronounced seasonal shifts in macronutrient and mineral composition, with the effect more pronounced in sheep milk than goat milk [1]. Compositional differences are also apparent between breeds within a single species: a study comparing two Chinese dairy goat breeds found significant differences in fatty-acid and triacylglycerol profiles depending on breed [2]. Mixing milk from different species is likewise reported to change the physicochemical and sensory properties of the resulting dairy products, which is relevant to regions where blended milk is common practice [3].

Buffalo milk, one of the four sources considered in this report, is consistently reported in the literature to differ from cow milk in ways that are relevant to a physicochemical comparison: a recent comprehensive review reports that buffalo milk has roughly twice the fat content of cow milk along with a higher protein and total-solids content, larger fat

globules, and a lower freezing point, and links these differences to buffalo milk’s greater viscosity and suitability for products such as ghee and paneer [12].

Processing choices also shape milk stability and shelf life. A comparison of high-pressure processing against conventional pasteurization in cow and goat milk found that both methods achieved a similar microbial shelf life under refrigeration, though goat milk showed greater susceptibility to spoilage by the end of the storage period [4]. Feeding system is another documented factor: pasture (“grass-fed”) feeding has been associated with a more favourable fatty-acid profile in bovine milk relative to non-pasture systems [5].

Milk spoilage in commercial supply chains is largely driven by microbial contamination. Two well-documented pathways are post-pasteurization contamination, in which bacteria such as *Pseudomonas* re-enter the milk after heat treatment, and the survival of psychrotolerant spore-forming bacteria that germinate under refrigeration [6, 7]. Reviews of dairy spoilage and food-loss data likewise point to microbial activity, rather than a single cause, as the dominant driver of shelf-life loss across the supply chain [8]. It is worth noting explicitly that none of these spoilage mechanisms were measured directly in this report; the dataset used here (Chapter 2) is a snapshot of physicochemical variables and does not include time-series or microbiological measurements, so statements about spoilage in this report are drawn from this literature rather than from original experimental evidence.

Taken together, the literature indicates that milk quality and stability depend on an interacting set of factors, species, feeding regime, processing method, and storage temperature, rather than any one variable in isolation. This motivates the correlation-based approach used later in this report to examine how quality indicators move together within the constructed dataset, while treating any spoilage-related claims as literature-derived context rather than as findings of this report’s own analysis.

Chapter 2

Methodology

This chapter documents how the dataset used in this report was constructed and analyzed. It is included in response to feedback that the original draft moved directly from a description of project activities to results, without making the dataset’s provenance, preprocessing, and analytical approach explicit.

2.1 Dataset Provenance and Construction

The dataset used in this report is **not a set of laboratory-measured milk samples**. It was synthetically constructed for this project using representative physicochemical value ranges drawn from publicly available dairy-science literature and open milk-quality datasets, rather than being collected through original laboratory testing. This is an important distinction for interpreting the results in Chapter 3: the dataset is intended to be representative of typical physicochemical relationships reported in the literature, not a measured sample of milk actually sold in Karnataka.

The construction pipeline followed these stages:

1. **Literature review** of typical physicochemical ranges (fat, SNF, protein, pH, lactose, density, freezing point, conductivity) reported for cow, buffalo, goat, and mixed milk.
2. **Definition of representative ranges** for each variable, by source, informed by the values reported in the reviewed literature (see Section 1.2 and the bibliography).
3. **Generation of a synthetic base dataset** by sampling within these representative ranges for each of the four source categories.
4. **Exploratory data analysis, correlation analysis, and visualization**, described in Sections 2.3–2.4 and reported in Chapter 3.

2.2 Dataset Description

The dataset is organized by milk source, **Cow, Buffalo, Goat, and Mixed**, with the variables summarized in Table 2.1.

Table 2.1: Variables recorded in the constructed dataset

Variable	Unit	Description
Source	—	Milk source category: Cow, Buffalo, Goat, or Mixed
Fat	%	Fat content as a percentage of total milk volume
SNF	%	Solids-not-fat: total solids excluding fat
Lactose	%	Lactose (milk sugar) content
Protein	%	Total protein content
Density	kg/m ³	Mass per unit volume of the milk sample
pH	—	Acidity/alkalinity on the standard pH scale
Added Water	%	Estimated proportion of added (diluting) water
Antibiotic Presence	Binary	Whether antibiotic residue is flagged as present
Freezing Point	°C	Freezing point, used as an adulteration indicator
Conductivity	mS/cm	Electrical conductivity of the milk sample
Classification	Categorical	Standard, Low Quality, or Fermented Milk

2.3 Data Preprocessing

Before analysis, the following preprocessing steps were applied:

- Standardizing column names and data types (numeric variables cast to float, Source and Classification cast to categorical).
- Checking for and handling missing or out-of-range values (see the placeholder in Section 2.2 for exact figures).
- Encoding the categorical Source and Classification variables for use in grouped comparisons and, where relevant, as model inputs.

2.4 Exploratory Analysis and Statistical Approach

Exploratory data analysis consisted of summary statistics by source group, followed by a Pearson correlation analysis across the numeric physicochemical variables, visualized as a heatmap (Section 3.2). Group-wise comparisons across the four milk sources (Section 3.3) were made using descriptive statistics (means and general spread) rather than formal hypothesis tests.

A limitation is noted explicitly here: the correlation coefficients reported in Chapter 3 are descriptive. Formal statistical significance testing (e.g. p-values for each correlation, confidence intervals, or a t-test / ANOVA comparing means across source groups) was not performed in this iteration of the analysis. Readers should not treat the reported correlation strengths as evidence of statistical significance without that additional testing, and it is flagged in Chapter 4 as a direction for future work.

2.5 Ethical Statement

No human or animal experimentation was conducted for this project. The dataset is synthetically constructed from published literature values and open data rather than from biological samples, so no biosafety, animal-welfare, or human-subjects approval was required.

Chapter 3

Results and Discussion

3.1 Activity Details

The service-learning project combined background research, dataset construction, and exploratory analysis. The initial phase focused on reviewing secondary literature on milk composition and spoilage to identify representative value ranges for the four milk sources (Chapter 2). The dataset was then explored using descriptive statistics, correlation analysis, and a supplementary classification exercise, with the findings compiled into this report.

The project gave practical experience in constructing and documenting an analysis-ready dataset, applying introductory statistical methods to it, and connecting the resulting patterns back to real-world dairy industry practice, while learning firsthand why transparent documentation of a dataset’s provenance matters for how its results should be interpreted.

3.2 Correlation Analysis

A Pearson correlation analysis was run across the numeric physicochemical variables in the dataset and visualized as a heatmap.

Within this dataset:

- **Fat and SNF:** a moderate positive correlation ($r = 0.53$) was observed, consistent with the general dairy-science expectation that fat and non-fat solids tend to move together across milk sources.
- **pH and SNF:** a moderate correlation ($r = 0.65$) was observed; lower pH in the dataset tended to co-occur with lower SNF.
- **Added Water and Density/Nutrient content:** a negative correlation was observed, consistent with the expected dilution effect of added water.

As noted in Section 2.4, these are descriptive correlation coefficients; no significance testing was performed, so they should be read as patterns observed in this particular dataset rather than as statistically confirmed relationships. It is also worth being explicit that correlation does not establish causation: for example, the pH–SNF correlation is consistent with, but does not by itself demonstrate, a causal link between acidification and solids content; both variables could instead be jointly influenced by milk source or processing stage.

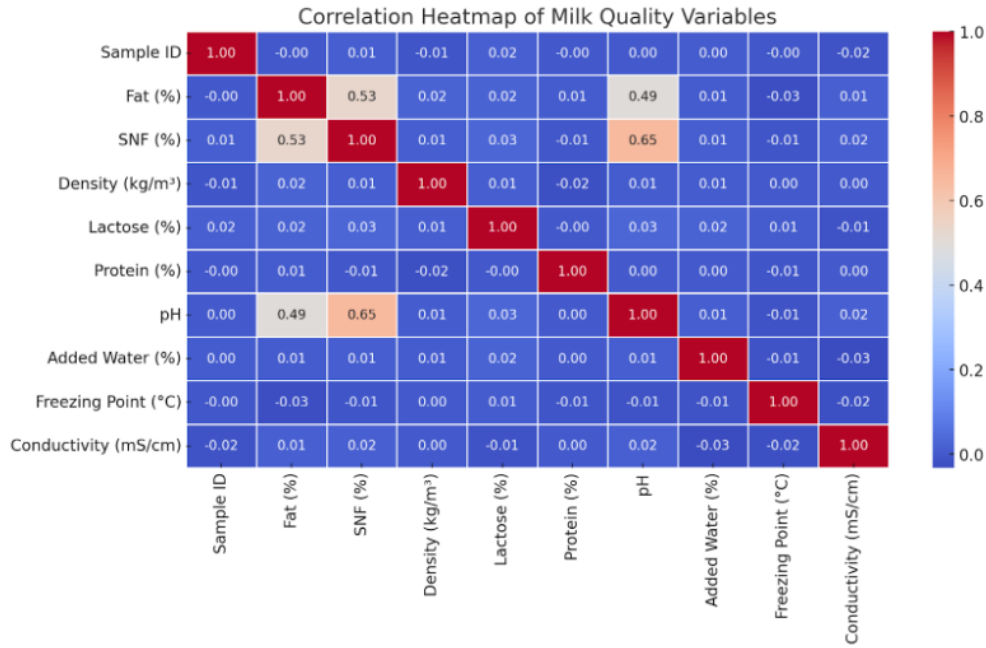


Figure 3.1: Pearson Correlation Matrix of Physicochemical Milk Quality Variables

3.3 Comparison Across Milk Sources

Comparing the four milk sources within the dataset showed the following descriptive patterns:

- **Buffalo milk** showed the highest fat and protein content among the four sources in the dataset, consistent with the literature, which reports buffalo milk fat content roughly double that of cow milk [12].
- **Cow milk** showed comparatively stable pH and moderate fat content relative to buffalo and goat milk.
- **Goat milk** showed lower fat but higher protein and lactose than cow milk, and a slightly lower pH.
- **Mixed milk** fell between the single-source categories, consistent with it being modelled as a blend.

On spoilage specifically: this dataset does not include time-series or microbiological measurements, so it cannot directly show which milk type spoils faster. The literature reviewed in Chapter 1 reports that lower-pH milk (such as goat milk, which tends to have a slightly lower pH than cow or buffalo milk) is generally more prone to faster souring, and that buffalo milk's higher fat and total-solids content is linked in the literature to greater viscosity and, in some studies, to a lower freezing point [12, 4]. These are presented here as literature-derived context for interpreting the pH differences observed across sources in the dataset, not as findings this report's own data can demonstrate.

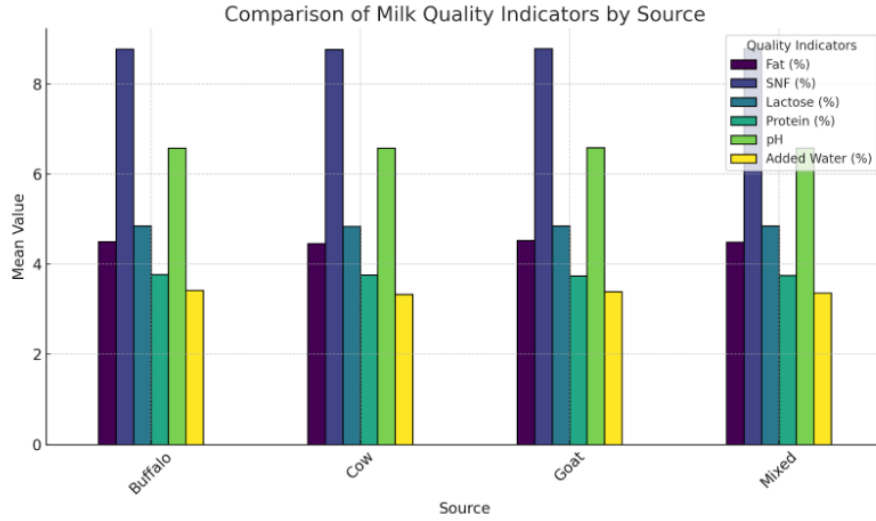


Figure 3.2: Comparison of Physicochemical Milk Quality Indicators by Source (Cow, Buffalo, Goat, Mixed)

3.4 Supplementary Classification Analysis

As a supplementary exercise, a Random Forest classifier was trained on the physicochemical variables to predict the dataset’s Classification label (Standard / Low Quality / Fermented). The aim of this exercise was exploratory: to check whether the recorded physicochemical variables carry enough signal to separate the three classes, as a complement to the descriptive correlation and comparison analysis above, rather than to build a production-ready quality-grading model.

3.5 Relevance to Dairy Industry Practice

The variables considered in this report, particularly added-water percentage, freezing point, and pH, correspond to indicators that Indian dairy regulation already monitors in practice. The Food Safety and Standards Authority of India (FSSAI) sets compositional and adulteration-related standards for milk sold in India, and dairy processors more broadly are expected to follow Hazard Analysis and Critical Control Points (HACCP)-based food-safety management to control microbial and chemical risk through processing and storage. The patterns discussed in this report, for example, the use of freezing point and density as adulteration indicators, are consistent with the kinds of checks these frameworks already require, though this report does not evaluate compliance with either framework directly.

3.6 Scope of the Work and Future Directions

This study offers a starting point, using a synthetically constructed but literature-grounded dataset, for further work on:

```
• Click to add a breakpoint rning algo. takes 5000 sample, and trained on those. based on

import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
from sklearn.metrics import accuracy_score, classification_report

# Assign categories based on conditions
def classify_milk(row):
    if row["Aflatoxin M1 (ppb)"] > 0.04:
        return "Green (Low Quality)"
    elif row["Fat (%)"] >= 4.5 and row["SNF (%)"] >= 9.0 and row["pH"] >= 6.6:
        return "Orange (Premium Quality)"
    elif row["pH"] < 6.5:
        return "Curd (Best for Fermentation)"
    else:
        return "Blue (Standard Quality)"

df["Milk Category"] = df.apply(classify_milk, axis=1)

# Encode the categorical target variable (Milk Category)
label_encoder = LabelEncoder()
df["Milk Category Encoded"] = label_encoder.fit_transform(df["Milk Category"])

# Define features and target variable
X = df[["Fat (%)", "SNF (%)", "pH", "Aflatoxin M1 (ppb)"]] # Features
y = df["Milk Category Encoded"] # Target

# Normalize features
scaler = MinMaxScaler()
X = scaler.fit_transform(X)
```

Figure 3.3: Random Forest classification code

- Replacing or supplementing the synthetic dataset with real, laboratory-measured samples from Karnataka dairies, to test whether the patterns observed here hold in measured data.
- Adding formal statistical significance testing (p-values, confidence intervals) to the correlation and group-comparison results.
- Time-series or microbiological measurement of spoilage, which this report's snapshot dataset cannot address directly.
- A more rigorously evaluated classification or regression model, with cross-validation and reporting of per-class metrics, building on the exploratory Random Forest exercise in Section 3.4.

```

# Split into training (80%) and testing (20%) sets
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)

# Train a Random Forest classifier with optimized parameters
best_model = RandomForestClassifier(n_estimators=200, max_depth=20, min_samples_split=5, min_samples_leaf=2, random_state=42, n_jobs=-1)
best_model.fit(X_train, y_train)

# Predictions
y_pred = best_model.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
report = classification_report(y_test, y_pred, target_names=label_encoder.classes_)

# Print results
print(f"Optimized Model Accuracy: {accuracy * 100:.2f}%")
print("\nClassification Report:\n", report)

```

Figure 3.4: Python implementation of the data pipeline and Random Forest classifier

- Detection of adulteration using indicators such as pH, added-water percentage, and freezing point, benchmarked against FSSAI thresholds.
- The potential influence of seasonal and regional variation on milk composition and stability.
- The role emerging technologies, such as biosensors and spectroscopic methods, could play in faster milk-quality assessment.

Chapter 4

Conclusion

This report presented a descriptive statistical analysis of a synthetically constructed, literature-grounded dataset representing physicochemical milk-quality characteristics across cow, buffalo, goat, and mixed milk. The study demonstrated a transparent workflow for constructing a representative dataset from published literature, performing exploratory data analysis, examining relationships between key quality indicators through correlation analysis, comparing physicochemical characteristics across milk sources, and conducting a supplementary Random Forest classification exercise to explore whether the recorded variables could distinguish milk-quality categories. The observed patterns were interpreted alongside findings from the dairy-science literature, allowing the analytical results to be placed within a broader domain context.

Limitations: The primary limitation of this study is that the dataset was synthetically constructed from representative values reported in the literature rather than obtained through direct laboratory measurement. Consequently, the findings should be interpreted as a methodological demonstration and exploratory statistical analysis rather than as evidence describing the actual composition or quality of commercially available milk in Karnataka or elsewhere. Furthermore, no formal statistical significance testing (such as hypothesis tests or confidence intervals) was performed, and the dataset does not contain longitudinal or microbiological measurements required to directly investigate spoilage dynamics. Accordingly, all spoilage-related discussions are derived from published literature rather than from the dataset itself. Finally, the reported correlations should be interpreted as statistical associations and not as evidence of causal relationships between the physicochemical variables.

Future work: Building on the Scope section in Chapter 3, the most valuable next steps would be validating these patterns against real, laboratory-measured milk samples; adding formal significance testing; and extending the classification exercise with proper cross-validation and evaluation reporting.

Overall, this report is intended as a modest, transparently-documented contribution: a worked example of constructing and analyzing a representative milk-quality dataset, rather than a definitive empirical study of the dairy industry.

4.1 Experience and Learning Outcomes

Working on this project gave me hands-on experience with the full arc of a small applied-statistics project: constructing a dataset from literature-derived ranges, applying SMOTE to address class imbalance, running exploratory data analysis and Pearson correlation

analysis, and, at a more introductory level, thinking through what a proper hypothesis test would need to add rigor to the correlation results I was reporting. Using Python for this workflow (pandas for data handling, standard plotting libraries for the heatmap and comparison figures, and scikit-learn for the Random Forest exercise) let me move from raw variable ranges to a documented, analysis-ready dataset relatively efficiently.

Putting this report together was also a useful lesson in the importance of documenting a dataset's provenance clearly. Realizing, in revising this report, that the difference between a synthetic and a measured dataset changes how every downstream claim should be phrased was probably the single most useful methodological lesson from this project, more so than any individual statistical technique.

Beyond the technical workflow, background reading on dairy industry practice, and discussion with people familiar with milk processing, gave me useful context for interpreting the dataset and for understanding, at a general level, the kinds of testing and regulatory standards (such as FSSAI norms) that real dairy producers work within.

Overall, this project strengthened my practical skills in exploratory data analysis, correlation-based statistical reasoning, and structuring a Python data-analysis workflow, and gave me a clearer sense of the gap between an exploratory academic exercise and a validated, industry-ready analysis.

Appendix A

Dataset and Code

This appendix contains supporting screenshots referenced from Chapter 3: a spreadsheet view of the constructed dataset, the Random Forest classification output, and the Python implementation. Table 2.1 in Section 2.2 is the primary reference for variable definitions; the screenshot below is included only as supporting documentation of the underlying data.

Sample ID	State	District	Source	Fat (%)	SNF (%)	Density (kg/m³)	Lactose (%)	Protein (%)	pH	Added Water (%)	Antibiotic Presence	Freezing Point (°C)	Conductivity (mS/cm)	Milk Category
1000	Karnataka	Hubballi-D	Goat	4.6	9	1027	4.9	3.9	6.5	0	Yes	-0.55	4.4	Curd (Best for Fermentation)
1001	Karnataka	Dakshina	Cow	3.9	9.2	1035	4.6	4.1	6.6	2	No	-0.51	4.2	Blue (Standard Quality)
1002	Karnataka	Gulbarga	P	3.9	9.2	1034	4.8	4.3	6.7	5	No	-0.54	4	Green (Low Quality)
1003	Karnataka	Ballari	Mixed	4.8	9	1031	4.5	3.2	6.6	7	Yes	-0.51	4.1	Curd (Best for Fermentation)
1004	Karnataka	Belagavi	Cow	4.7	8.6	1033	5.2	3.4	6.6	0	Yes	-0.52	4.4	Blue (Standard Quality)
1005	Karnataka	Ballari	Mixed	4.8	8.7	1031	5.2	3.3	6.3	7	No	-0.51	4.1	Curd (Best for Fermentation)
1006	Karnataka	Vijayapura	Goat	5.3	8.9	1032	4.8	4.1	6.6	7	Yes	-0.52	4.2	Curd (Best for Fermentation)
1007	Karnataka	Vijayapura	Mixed	5.2	9.2	1031	4.5	4	6.5	3	Yes	-0.53	4	Curd (Best for Fermentation)
1008	Karnataka	Ballari	Buffalo	3.9	8.9	1029	4.5	4.1	6.4	3	No	-0.55	4.3	Curd (Best for Fermentation)
1009	Karnataka	Vijayapura	Mixed	3.9	8.6	1025	4.9	4.4	6.6	0	Yes	-0.53	4.1	Curd (Best for Fermentation)
1010	Karnataka	Bengaluru	Buffalo	4.2	8.6	1029	5	3.6	6.5	5	No	-0.53	4.2	Curd (Best for Fermentation)
1011	Karnataka	Udupi	Goat	5.4	8.9	1035	4.8	3.3	6.8	3	Yes	-0.53	4.4	Green (Low Quality)
1012	Karnataka	Mandya	Mixed	4.8	8.8	1025	4.9	3.8	6.6	5	No	-0.5	4	Curd (Best for Fermentation)
1013	Karnataka	Dakshina	Mixed	3.7	9.1	1033	4.9	3.4	6.8	3	Yes	-0.52	4.4	Green (Low Quality)
1014	Karnataka	Chikmagalur	Buffalo	5.1	8.5	1032	5	3.9	6.6	7	Yes	-0.54	4.2	Blue (Standard Quality)
1015	Karnataka	Bagalgot	Buffalo	4.8	9	1034	4.6	4.1	6.8	3	No	-0.51	4.4	Green (Low Quality)
1016	Karnataka	Gulbarga	Buffalo	3.6	8.7	1026	4.6	4.4	6.7	3	No	-0.53	4.4	Green (Low Quality)
1017	Karnataka	Kodagu	Goat	5.5	9.1	1034	5.1	3.3	6.6	0	No	-0.55	4.3	Curd (Best for Fermentation)
1018	Karnataka	Gadag	Goat	4.8	8.8	1028	4.9	3.8	6.6	0	No	-0.53	4.4	Curd (Best for Fermentation)
1019	Karnataka	Chamarajji	Cow	3.7	8.9	1033	4.7	3.2	6.6	7	No	-0.54	4.2	Blue (Standard Quality)
1020	Karnataka	Mandya	Goat	5.9	9.3	1029	5.1	3.8	6.8	3	Yes	-0.53	4.3	Green (Low Quality)
1021	Karnataka	Gadag	Mixed	3.8	8.5	1030	4.8	4.4	6.6	0	No	-0.54	4.3	Curd (Best for Fermentation)
1022	Karnataka	Tumakuru	Buffalo	4.1	8.5	1029	4.7	4.3	6.6	5	Yes	-0.51	4.1	Curd (Best for Fermentation)
1023	Karnataka	Shivamogga	Cow	5.8	9.3	1026	4.5	4.5	6.6	7	No	-0.52	4.4	Orange (Premium Quality)
1024	Karnataka	Davanageri	Mixed	4.3	8.9	1033	4.8	3.3	6.7	7	Yes	-0.54	4.4	Green (Low Quality)
1025	Karnataka	Mysuru	Mixed	3.6	9	1034	4.5	3.6	6.8	2	No	-0.5	4.4	Green (Low Quality)
1026	Karnataka	Vijayapura	Mixed	4.9	8.8	1029	4.6	3.6	6.6	7	No	-0.53	4.3	Curd (Best for Fermentation)
1027	Karnataka	Bengaluru	Mixed	5.4	8.9	1034	5	3.2	6.7	7	No	-0.53	4.2	Green (Low Quality)
1028	Karnataka	Mysuru	Mixed	2.5	7.9	1034	4.9	4.3	6	5	No	-0.51	4.1	Green (Low Quality)

Figure A.1: Spreadsheet view of the constructed dataset (see Table 2.1 for variable definitions)

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